



JABchem



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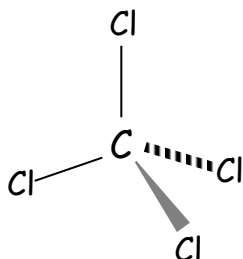
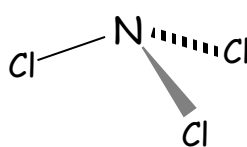
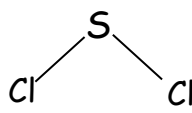
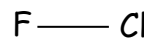
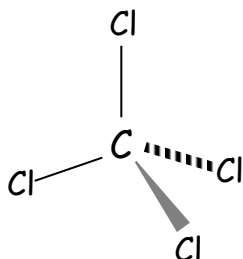
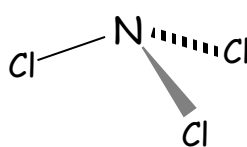
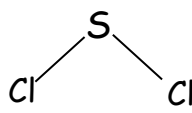
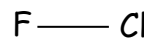
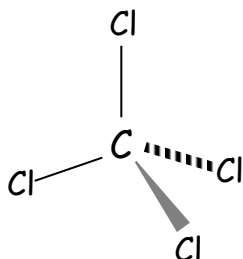
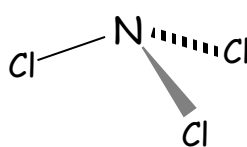
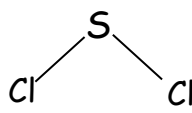
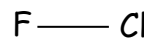
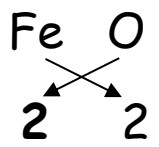
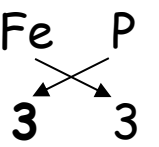
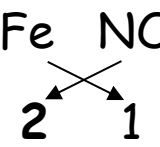
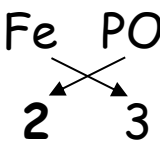
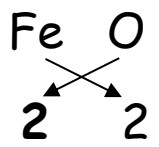
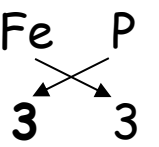
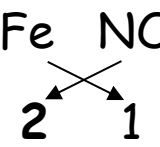
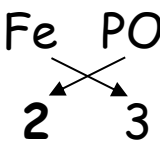
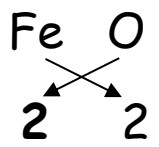
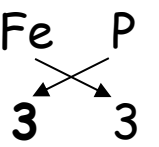
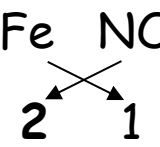
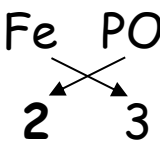
Past Papers Nat 5 Chemistry

2023 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/100)	%	
A	69+	69%	42.5%
B	57+	57%	20.3%
C	46+	46%	15.1%
D	34+	34%	12.2%
No award	<34	<34%	9.9%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.1 /25	46.0 /75	No Assignment in 2023

2023 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning																
1	D	62	$\text{Rate} = \frac{\Delta \text{Quantity}}{\Delta \text{Time}} = \frac{300 \text{ cm}^3}{60 \text{ s}} = 5 \text{ cm}^3 \text{ s}^{-1}$																
2	A	92	<table border="1"> <thead> <tr> <th>Particle</th><th>Location</th><th>Charge</th><th>Mass</th></tr> </thead> <tbody> <tr> <td>proton</td><td>inside nucleus</td><td>+1</td><td>1 amu</td></tr> <tr> <td>neutron</td><td>inside nucleus</td><td>0</td><td>1 amu</td></tr> <tr> <td>electron</td><td>outside nucleus</td><td>-1</td><td>approx 0</td></tr> </tbody> </table>	Particle	Location	Charge	Mass	proton	inside nucleus	+1	1 amu	neutron	inside nucleus	0	1 amu	electron	outside nucleus	-1	approx 0
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3	C	80	<table border="1"> <thead> <tr> <th>A</th><th>B</th><th>C</th><th>D</th></tr> </thead> <tbody> <tr> <td> CCl_4  </td><td> NCl_3  </td><td> SCl_2  </td><td> FCl  </td></tr> <tr> <td>tetrahedral</td><td>trigonal pyramidal</td><td>angular</td><td>linear</td></tr> </tbody> </table>	A	B	C	D	CCl_4 	NCl_3 	SCl_2 	FCl 	tetrahedral	trigonal pyramidal	angular	linear				
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CCl_4 	NCl_3 	SCl_2 	FCl 																
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4	B	58	Lithium Chloride is ionic as compound has metal bonded to non-metal. ☒ A Diagram shows metallic bonding (positive ions surrounded by delocalised electrons) ☒ B Diagram shows ionic bonding (positive ions and negative ions) ☒ C Diagram shows covalent molecular (separate molecules with covalent bonds inside molecule) ☒ D Diagram shows covalent network (long network of covalent bonds)																
5	B	52	<table border="1"> <thead> <tr> <th>A</th><th>B</th><th>C</th><th>D</th></tr> </thead> <tbody> <tr> <td> Fe O  FeO </td><td> Fe P  FeP </td><td> Fe NO_3^-  $\text{Fe}(\text{NO}_3)_2$ </td><td> Fe PO_4^{3-}  $\text{Fe}_3(\text{PO}_4)_2$ </td></tr> <tr> <td>valency of Fe = 2</td><td>valency of Fe = 3</td><td>valency of Fe = 2</td><td>valency of Fe = 2</td></tr> </tbody> </table>	A	B	C	D	Fe O  FeO	Fe P  FeP	Fe NO_3^-  $\text{Fe}(\text{NO}_3)_2$	Fe PO_4^{3-}  $\text{Fe}_3(\text{PO}_4)_2$	valency of Fe = 2	valency of Fe = 3	valency of Fe = 2	valency of Fe = 2				
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valency of Fe = 2	valency of Fe = 3	valency of Fe = 2	valency of Fe = 2																
6	D	71	☒ A no. of mol = volume x concentration = $0.1 \text{ litres} \times 1.00 \text{ mol l}^{-1} = 0.1 \text{ mol}$ ☒ B no. of mol = volume x concentration = $0.15 \text{ litres} \times 0.75 \text{ mol l}^{-1} = 0.1125 \text{ mol}$ ☒ C no. of mol = volume x concentration = $0.2 \text{ litres} \times 0.60 \text{ mol l}^{-1} = 0.12 \text{ mol}$ ☒ D no. of mol = volume x concentration = $0.25 \text{ litres} \times 0.25 \text{ mol l}^{-1} = 0.0625 \text{ mol}$																
7	B	54	☒ A aluminium oxide is insoluble $\therefore$ no change to pH of water ☒ B barium oxide is a soluble metal oxide $\therefore$ dissolves in water to form alkaline pH>7 ☒ C nitrogen oxide is a non-metal oxide $\therefore$ dissolves in water to form pH<7 ☒ D hydrogen oxide H_2O is water with a pH=7 $\therefore$ does not change pH of water when added																
8	C	45	nickel carbonate + sulfuric acid $\longrightarrow$ nickel sulfate + water + carbon dioxide nickel hydroxide + sulfuric acid $\longrightarrow$ nickel sulfate + water nickel + sulfuric acid $\longrightarrow$ nickel sulfate + hydrogen ☒ A nickel hydroxide does not produce a gas when reacted with sulfuric acid ☒ B nickel does not produce a gas when reacted with sulfuric acid ☒ C nickel sulfate is produced in all three reactions ☒ D no water produced in reaction of nickel and sulfuric acid $\therefore$ not a neutralisation reaction																

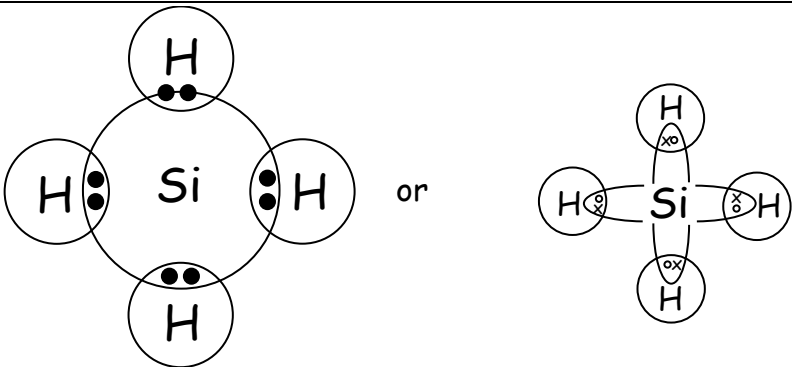
9	B	59	$(\text{Na}^+)_2\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+\text{Cl}^-(\text{aq}) \rightarrow 2\text{Na}^+\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ Split solutions into ions $2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ Identify Spectator Ions and cancel out Spectator Ions $2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ Re-write equation without spectator ions $\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$															
10	A	49	Carbon number: 5 4 3 2 1 $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{C}(\text{CH}_3)_2\text{CH}_3$ ↑ 1x methyl group on C₄ ↑ 2x methyl group on C₂ Name of compound: 2,2,4-trimethylpentane NB: Overall formula is of alkane as formula C₈H₁₈ fits general formula C_nH_{2n+2}															
11	C	56	☒A This structure is also 2-methylbut-2-ene but drawn differently ☒B This structure has formula C₅H₁₂ but 2-methylbut-2-ene has formula C₅H₁₀ ☒C Both have formula C₅H₁₀ and have different structures ∴ isomers ☒D This structure has formula C₆H₁₂ but 2-methylbut-2-ene has formula C₅H₁₀															
12	B	33	☒A Addition of H₂ across the C=C double bond in but-2-ene produces butane C₄H₁₀ ☒B Butan-1-ol cannot be produced as -OH group must be added to C₂ to form butan-2-ol ☒C Addition of H₂O across the C=C double bond in but-2-ene produces butan-2-ol ☒D Addition of Br₂ across the C=C double bond in but-2-ene produces 2,3-dibromobutane															
13	A	87	O C H₅C₂ OH + O C C₅H₁₁ HO → O C H₅C₂ C₅H₁₁ + CO₂ + H₂O															
14	D	68	☒A methanol CH₃OH has a lower formula mass compared to octan-1-ol C₈H₁₇OH ☒B CH₃OH has a higher solubility than C₈H₁₇OH as methanol has a shorter carbon chain ☒C methanol CH₃OH has a lower formula mass compared to octan-1-ol C₈H₁₇OH ☒D methanol CH₃OH has a lower formula mass and higher solubility than octan-1-ol C₈H₁₇OH															
15	D	78	<table><tr><th>Chemistry</th><th>butane</th><th>cyclobutane</th></tr><tr><td>General Formula</td><td>C₄H₁₀ fits C_nH_{2n+2}</td><td>C₄H₈ fits C_nH_{2n}</td></tr><tr><td>Melting Point</td><td>-138°C</td><td>-90°C</td></tr><tr><td>Solubility in water</td><td>insoluble in water</td><td>insoluble in water</td></tr><tr><td>Saturation</td><td>Saturated (no C=C double bonds)</td><td>Saturated (no C=C double bonds)</td></tr></table>	Chemistry	butane	cyclobutane	General Formula	C ₄ H ₁₀ fits C _n H _{2n+2}	C ₄ H ₈ fits C _n H _{2n}	Melting Point	-138°C	-90°C	Solubility in water	insoluble in water	insoluble in water	Saturation	Saturated (no C=C double bonds)	Saturated (no C=C double bonds)
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16	C	69	Acid + Metal Oxide → Salt + Water methanoic acid + sodium oxide → sodium methanoate + water															
17	A	72	☒A conducts as solid and liquid ∴ metallic bonding ☒B does not conduct as solid or liquid and has low m.pt. and b.pt. ∴ covalent molecular ☒C conducts as liquid but not solid ∴ ionic bonding ☒D does not conduct as solid or liquid and has high m.pt. ∴ covalent network															
18	B	70	☒A Y is least reactive as it is only metal not to react with acid. (Least reactive comes first in list) ☒B Y is least reactive and Z is most reactive ∴ Y then X then Z ☒C Z is most reactive as it is only metal to react with water. (Most reactive comes last in list) ☒D Z is most reactive as it is only metal to react with water. (Most reactive comes last in list)															

19	D	86	☒ A Nickel(A) is lower than zinc(B) in Electrochemical series so electrons flow from B to A ☒ B Zinc(A) is lower than aluminium(B) in Electrochemical series so electrons flow from B to A ☒ C Aluminium(A) is lower than magnesium(B) in Electrochemical series so electrons flow from B to A ☒ D Aluminium(A) is higher than nickel(B) in Electrochemical series so electrons flow from A to B			
20	A	29	As Magnesium is higher up the electrochemical series. Magnesium will oxidise into Mg^{2+} ions and Ag^+ ions will reduce back to silver atoms in a redox reaction. Oxidation $Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$ Reduction $Ag^+(aq) + e^- \longrightarrow Ag(s)$ Oxidation $Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$ Reduction x2 $2Ag^+(aq) + 2e^- \longrightarrow 2Ag(s)$ Redox $Mg(s) + 2Ag^+(aq) \longrightarrow Mg^{2+}(aq) + 2Ag(s)$			
21	C	72	Process	Reactants	Products	Catalyst
			Haber	Nitrogen + Hydrogen	Ammonia	Iron
			Ostwald	Ammonia + oxygen	Nitric acid (ultimately)	Platinum
22	C	74	${}_{90}^{227}Th \xrightarrow{\alpha} {}_{88}^{223}Ra \xrightarrow{\alpha} {}_{86}^{219}Rn \xrightarrow{\alpha} {}_{84}^{215}Po \xrightarrow{\alpha} {}_{82}^{211}Pb$			
23	C	70	☒ A Barium sulfate is insoluble and can be collected in a precipitation reaction by filtration ☒ B Lead(II) sulfate is insoluble and can be collected in a precipitation reaction by filtration ☒ C Calcium chloride is soluble so will not form a precipitate ☒ D Silver Chloride is insoluble and can be collected in a precipitation reaction by filtration			
24	A	80	☒ A Glucose gives blue to orange with Benedict's and sodium chloride gives yellow flame test ☒ B Starch is not present as there was no change in the iodine test ☒ C Flame test gave yellow flame but strontium chloride would give a red flame test result ☒ D Starch is not present as there was no change in the iodine test			
25	B	44	$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$ 2mol 0.004mol 1mol 0.002mol			

2023 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning																
1a	7	<table><tr><td>Diatomic Element</td><td>Hydrogen</td><td>Nitrogen</td><td>Oxygen</td><td>Fluorine</td><td>Chlorine</td><td>Bromine</td><td>Iodine</td></tr><tr><td>Formula</td><td>H₂</td><td>N₂</td><td>O₂</td><td>F₂</td><td>Cl₂</td><td>Br₂</td><td>I₂</td></tr></table>	Diatomic Element	Hydrogen	Nitrogen	Oxygen	Fluorine	Chlorine	Bromine	Iodine	Formula	H ₂	N ₂	O ₂	F ₂	Cl ₂	Br ₂	I ₂
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Formula	H ₂	N ₂	O ₂	F ₂	Cl ₂	Br ₂	I ₂											
1b	35	Chlorine 35 is the common isotope in sample as the average is closer to 35 than 37																
1c	One from: <table><tr><td>Fluorine</td><td>Bromine</td></tr><tr><td>Iodine</td><td>Astatine</td></tr></table>	Fluorine	Bromine	Iodine	Astatine	Elements in the same chemical group have similar chemical properties due to having the same number of outer electrons.												
Fluorine	Bromine																	
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1d	<table><tr><td>10</td><td></td></tr><tr><td></td><td>20</td></tr></table>	10			20	<table><tr><td>²⁴₁₂Mg²⁺</td><td>No. of protons = atomic number = 12 No of neutrons = mass number - atomic number = 24 - 12 = 12 No of electrons = atomic number - charge = 12 - (2) = 10</td></tr><tr><td>³⁷₁₇Cl⁻</td><td>No. of protons = atomic number = 17 No of neutrons = mass number - atomic number = 37 - 17 = 20 No of electrons = atomic number - charge = 17 - (-1) = 18</td></tr></table>	²⁴ ₁₂ Mg ²⁺	No. of protons = atomic number = 12 No of neutrons = mass number - atomic number = 24 - 12 = 12 No of electrons = atomic number - charge = 12 - (2) = 10	³⁷ ₁₇ Cl ⁻	No. of protons = atomic number = 17 No of neutrons = mass number - atomic number = 37 - 17 = 20 No of electrons = atomic number - charge = 17 - (-1) = 18								
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2a	Hydrocarbon	A hydrocarbons is a compounds containing carbon and hydrogen only.																
2b(i)	Hydrogen	C ₂ H ₆ → C ₂ H ₄ + H ₂																
2b(ii)	decolourises bromine solution	Unsaturated compounds will decolourise bromine solution quickly. Bromine adds across the double bond in ethene to form 1,2-dibromoethane																
2c(i)	H-C≡C-H	Carbon has valency of 4 and makes four bonds per carbon atom Hydrogen has valency of 1 and make only one bond per hydrogen atom																
2c(ii)A	carbon dioxide and water	Ethyne is a hydrocarbon containing only carbon and hydrogen atoms. - Complete combustion of carbon produced carbon dioxide. - Complete combustion of hydrogen produced water.																
2c(ii)B	Gives out heat	Exothermic reactions are reactions where heat is given out to the surroundings. Endothermic reactions are reactions where heat is taken in from the surroundings																
3a	Group 2	Only Groups 2 contains metals only. Group 1 starts with the non-metal hydrogen. The remaining groups 3-0 all start with a non-metal at the top of the group																
3b(i)	KMnO ₄	<table><tr><td>Write down Symbols and valency below</td><td>Cross-Over arrows to work out formula</td><td>Work out chemical formula (Cancel down if necessary)</td></tr><tr><td> K MnO₄⁻ 1 1 </td><td> K MnO₄⁻ 1 1 </td><td>KMnO₄</td></tr></table>	Write down Symbols and valency below	Cross-Over arrows to work out formula	Work out chemical formula (Cancel down if necessary)	K MnO₄⁻ 1 1	K MnO₄⁻ 1 1	KMnO ₄										
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3b(ii)	Aluminium or zinc	<table><tr><td>Observation</td><td>Blinding white light</td><td>Bright light</td><td>Red glow with few sparks</td><td>Dull red glow</td></tr><tr><td>Metal</td><td>Magnesium</td><td>X</td><td>Iron</td><td>Copper</td></tr><tr><td>Reactivity</td><td colspan="4">High ▶ Low</td></tr></table> Conclusion: X must have reactivity between Magnesium and Iron ∴ 2 possible metals: aluminium or zinc	Observation	Blinding white light	Bright light	Red glow with few sparks	Dull red glow	Metal	Magnesium	X	Iron	Copper	Reactivity	High ▶ Low				
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3c	Hydrogen Burns with a pop	<table><tr><td>Gas</td><td>Hydrogen</td><td>Oxygen</td><td>Carbon Dioxide</td></tr><tr><td>Test</td><td>Burns with a pop</td><td>Relights a glowing splint</td><td>Turns limewater cloudy</td></tr></table>	Gas	Hydrogen	Oxygen	Carbon Dioxide	Test	Burns with a pop	Relights a glowing splint	Turns limewater cloudy								
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3d(i)	Voltage between 0.5 and 2.7	As Iron is between magnesium and tin on the electrochemical series, the voltage of an iron/copper cell will be between the voltage of a magnesium/copper cell and a tin/copper cell																
3d(ii)	Electrically conducting solutions containing ions	An electrolyte is an ion containing solution which completes the circuit in a cell and allows electricity to flow.																

3d(iii)	one answer from:	temperature	concentration	Type of electrode	Distance between electrodes	Surface area of electrodes																		
4a	From air	Problem Solving: Extracting information from a passage																						
4b	Lowers temperature or don't get used up	A catalyst speeds up a chemical reaction but does not get used up in the reaction. This often means that a chemical reaction can proceed at a lower temperature.																						
4c	534.1	1 litre jet fuel made from 4700g CO ₂ 5 litres jet fuel made from 5x4700g CO ₂ = 23500g CO gfm CO ₂ = (1x12)+(2x16) = 12+32 = 44g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{23500}{44} = 534.1\text{mol}$																						
5	Open Ended Question:	3 mark answer		2 mark answer		1 mark answer																		
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6a(i)	carboxyl group	<table><tr><td>—O—H hydroxyl group</td><td>$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$ carboxyl group</td></tr></table>					—O—H hydroxyl group	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$ carboxyl group																
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6a(ii)	Addition Polymerisation	C=C double bonds in monomer open up and join together to form polymer																						
6a(iii)	Polymer diagram:	<table><tr><td>$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ Monomer</td><td>$\begin{array}{c} \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \\ \quad \quad \quad \quad \quad \\ \text{—C—C—C—C—C—C—} \\ \quad \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$ Polymer</td><td>$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{—C—C—} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ Repeating Unit</td></tr></table>					$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ Monomer	$\begin{array}{c} \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \quad \text{H} \quad \text{COOH} \\ \quad \quad \quad \quad \quad \\ \text{—C—C—C—C—C—C—} \\ \quad \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$ Polymer	$\begin{array}{c} \text{H} \quad \text{COOH} \\ \quad \\ \text{—C—C—} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ Repeating Unit															
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6b(i)	CsCl and Material A	Problem Solving: Prediction of value and completion of bar graph																						
6b(ii)	Bar bigger than 18 and lower than 32	As strontium is the metal between calcium and barium in group 2, the prediction for SrCl ₂ would be between CaCl ₂ (18) and BaCl ₂ (32).																						
6c(i)	measuring cylinder or pipette	Beakers are less accurate than measuring cylinders at measuring volumes of liquid. Pipettes are the most accurate device for measuring exact volumes.																						
6c(ii)	Line Graph Showing:	<table><tr><td>1 mark</td><td>1 mark</td><td>1 mark</td><td>1 mark</td></tr><tr><td>One mark is awarded for a graph which shows points plotted rather than bars.</td><td>The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales </td><td>The axes of the graph have suitable labels and units.</td><td>All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.</td></tr></table>					1 mark	1 mark	1 mark	1 mark	One mark is awarded for a graph which shows points plotted rather than bars.	The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales	The axes of the graph have suitable labels and units.	All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.										
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7a	Trisilane	<table><tr><td>Prefix</td><td>Mono-</td><td>Di-</td><td>Tri-</td><td>Tetra-</td><td>Penta-</td><td>Hexa-</td><td>Hepta-</td><td>Octa-</td></tr><tr><td>Meaning</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>					Prefix	Mono-	Di-	Tri-	Tetra-	Penta-	Hexa-	Hepta-	Octa-	Meaning	1	2	3	4	5	6	7	8
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7b	Si ₅ H ₁₂	General Formula of silanes is Si _n H _{2n+2} in line with alkanes C _n H _{2n+2} For pentasilane, n=5 ∴ 2n+2 = (2x5)+2 = 12 ∴ Pentasilane formula = Si ₅ H ₁₂																						

7c	185	Compound	Monosilane	Disilane	Trisilane	Tetrasilane	Pentasilane	Hexasilane
		Formula	SiH ₄	Si ₂ H ₆	Si ₃ H ₈	Si ₄ H ₁₀	Si ₅ H ₁₂	Si ₆ H ₁₄
		Boiling Point (°C)	-112	-15	53	108	153	-
		Difference		97	68	55	45	(35)
		Prediction (°C)	-	-	-	-	-	185
7d	One diagram from:							
7e	Stronger intermolecular forces due to longer silicon chain	1st Mark: pentasilane has stonger/larger forces of attraction. 2nd Mark: forces of attractions are intermolecular/between molecules The longer the silicon chain, the greater the number of atoms within the molecule. Bigger molecules have stronger intermolecular bonds between molecules which raises the boiling point of pentasilane over tetrasilane						
7f	3.1	gfm SiO₂ = (1×28)+(2×16) = 28+32 = 60g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{6}{60} = 0.1\text{mol}$ $7\text{Mg} + 2\text{SiO}_2 + 14\text{HCl} \longrightarrow \text{Si}_2\text{H}_6 + 7\text{MgCl}_2 + 4\text{H}_2\text{O}$ 2mol 1mol 0.1mol 0.05mol gfm Si₂H₆ = (2×28)+(6×1) = 56+6 = 62g mass = no. of mol × gfm = 0.05mol × 62 = 3.1g						
8a	Fluorapatite	Problem Solving: extracting information from a passage						
8b	C ₆ H ₁₂ O	As there are no metals in the formula, any order of the non-metal elements is acceptable although most tend to list C then H then O.						
8c(i)	Nitrogen	ADP has the formula NH ₄ H ₂ PO ₄ contains nitrogen and phosphorus. Both are essential elements for plant growth. The 3rd essential element for plant growth is potassium.						
8c(ii)	Soluble	Fertilisers must contain at least one of the three essential elements nitrogen, phosphorus or potassium and be soluble in water.						
8d	31.6	gfm H₃PO₄ = (3×1)+(1×31)+(4×16) = 3+31+64 = 98g $\% \text{ P} = \frac{\text{mass of P}}{\text{gfm}} \times 100 = \frac{31}{98} \times 100 = 31.6\%$						
8e	Filtration	Solid calcium sulfate can be separated from a liquid by filtration.						
8f	$\frac{1}{2}$ mol or 0.5mol	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ 1mol : 2mol $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ 1mol : $\frac{1}{2}$mol						
9a(i)	0.627	heat energy = specific heat capacity × mass × change in Temperature $E_h = C \times m \times \Delta T$ $E_h = 4.18 \times 0.01 \times 15$ $E_h = 0.627 \text{ kJ}$						
9a(ii)	One from:	No draft shield	Energy lost to surroundings				Temperature not consistent in water	
		Glass absorbs heat	Incomplete combustion					

9b	150	1g biscuit releases 20.9 kJ 30g biscuit releases $20.9 \text{ kJ} \times \frac{30}{1}$ = 627 kJ 4.18kJ equals 1 kilocalorie 627 kJ equals $1 \text{ kilocalorie} \times \frac{627}{4.18}$ = 150 kilocalorie																																	
10a(i)	electrolysis	As caesium is more reactive than potassium then electrolysis would be the method to extract caesium from its ore. <table><tr><td>Method</td><td colspan="2">Electrolysis</td><td colspan="2">Heat With Carbon</td><td colspan="2">Heat Alone</td></tr><tr><td rowspan="3">Metals Made This Way</td><td>Potassium</td><td>Sodium</td><td>Zinc</td><td>Iron</td><td>Mercury</td><td>Silver</td></tr><tr><td>Lithium</td><td>Calcium</td><td>Nickel</td><td>Tin</td><td>Gold</td><td>Platinum</td></tr><tr><td>Magnesium</td><td>Aluminium</td><td>Lead</td><td>Copper</td><td></td><td></td></tr><tr><td>Reason</td><td colspan="2">most reactive metals</td><td colspan="2">medium reactive metals</td><td colspan="2">least reactive metals</td></tr></table>	Method	Electrolysis		Heat With Carbon		Heat Alone		Metals Made This Way	Potassium	Sodium	Zinc	Iron	Mercury	Silver	Lithium	Calcium	Nickel	Tin	Gold	Platinum	Magnesium	Aluminium	Lead	Copper			Reason	most reactive metals		medium reactive metals		least reactive metals	
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10a(ii)	Reduction	Metal ores contain metal ions which are reduced to produce metal atoms $\text{Cs}^+ + \text{e}^- \rightarrow \text{Cs}$																																	
10b(i)	$^0_{-1}\text{e}$	<table><tr><td>Particle</td><td>Proton</td><td>Neutron</td><td>Electron</td><td>Alpha</td><td>Beta</td></tr><tr><td>Symbol</td><td>^1_1p</td><td>^1_0n</td><td>$^0_{-1}\text{e}$</td><td>^4_2He</td><td>$^0_{-1}\text{e}$</td></tr></table>	Particle	Proton	Neutron	Electron	Alpha	Beta	Symbol	^1_1p	^1_0n	$^0_{-1}\text{e}$	^4_2He	$^0_{-1}\text{e}$																					
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11b(i)	$\text{WF}_6 + 3\text{H}_2\text{O} \rightarrow \text{WO}_3 + 6\text{HF}$	$\text{WF}_6 + 3\text{H}_2\text{O} \rightarrow \text{WO}_3 + 6\text{HF}$																																	
11b(ii)	concentration of hydrogen ions greater than concentration of hydroxide ions	<table><tr><td>Type</td><td colspan="2">Relationship between Concentration of Hydrogen ions and Hydroxide Ions</td></tr><tr><td>acid</td><td>Concentration of hydrogen ions greater than</td><td>Concentration of hydroxide ions</td></tr><tr><td>neutral</td><td>Concentration of hydrogen ions equal to</td><td>Concentration of hydroxide ions</td></tr><tr><td>alkali</td><td>Concentration of hydrogen ions less than</td><td>Concentration of hydroxide ions</td></tr></table>	Type	Relationship between Concentration of Hydrogen ions and Hydroxide Ions		acid	Concentration of hydrogen ions greater than	Concentration of hydroxide ions	neutral	Concentration of hydrogen ions equal to	Concentration of hydroxide ions	alkali	Concentration of hydrogen ions less than	Concentration of hydroxide ions																					
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11c	$\text{W}^{6+} + 2\text{e}^- \rightarrow \text{W}^{4+}$	When balancing an ion-electron equation, adding electrons will balance the charge by adding to the most positive side of the equation.																																	
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